

# AmazonHelp AI Support Agent

## 1. Problem

Build an AI customer-support agent for AmazonHelp that:

- classifies a customer message into a compact support-intent taxonomy,
- retrieves historical AmazonHelp interactions that show how similar issues were handled,
- drafts a grounded response using those historical resolutions,
- decides whether the request can be handled automatically or should go to a human.

The key design principle is that **retrieval relevance is not the same thing as safe autonomous answering**.

### What I chose not to build

I deliberately scoped this project as a grounded support-triage prototype rather than a full production customer-service platform. I did not build live order, account, payment, or delivery-system integrations because the assignment data does not provide authenticated access to those systems. I also did not build persistent multi-turn dialogue memory or claim full-fidelity multilingual support; those require additional product and language-specific evaluation. These were deferred so the project could focus on measurable intent classification, historical retrieval, grounded response generation, and conservative escalation.

## 2. Dataset and Golden Set

I selected AmazonHelp because its support corpus contains substantial multi-turn interactions across delivery, orders, refunds, returns, payments, accounts, membership, and digital products.

The evaluation uses a temporal split so evaluation conversations are not present in the retrieval corpus.

### Golden-set sampling and labeling

The 200-example review pool was constructed from the temporal-safe evaluation population with deliberate coverage of uncertain and boundary cases, rather than treating easy random examples as sufficient. Review priority was used to surface cases where intent or evidence decisions were more difficult, while the final primary benchmark retained only examples that received genuine human labels. A separate set of 48 model-assisted examples was preserved for exploration but excluded from primary ground truth.

The primary Gold evaluation contains **152 genuinely human-labelled examples**.

There are an additional 48 model-assisted examples in the repository, but these are explicitly treated as exploratory and are **not used as primary ground truth**.

## 3. Intent taxonomy

The final taxonomy contains 11 actionable support intents:

- delivery\_status
- delivery\_problem
- order\_issue
- preorder\_issue
- returns\_replacements
- refund\_issue
- payment\_billing
- account\_security
- prime\_membership
- digital\_product\_support
- general\_support

Annotation policy assigns the primary intent based on the customer's requested resolution. Escalation is modeled separately from intent.

## 4. System Architecture

### Intent classification

The final intent policy combines high-precision boundary rules with a MiniLM semantic fallback.

This performed better than:

- majority-class prediction,
- TF-IDF + Logistic Regression,
- pure MiniLM centroid classification,
- MiniLM kNN,
- LLM-only few-shot classification.

### Historical retrieval

I compared:

- BM25
- dense semantic retrieval
- hybrid BM25 + dense retrieval

The final system uses dense retrieval because stronger human/LLM quality measures favored semantic matches over broad lexical relatedness.

### Grounding

Retrieved interactions are evaluated for:

- evidence quality,
- actionability,
- grounding,
- whether autonomous answering is appropriate.

A dense similarity threshold is used only as an engineering pre-filter. It is **not treated as calibrated confidence**.

### Response generation

The response generator is implemented in src/generate.py.

It receives:

- the customer query,
- retrieved historical AmazonHelp interactions.

The prompt explicitly requires:

- historical evidence only,
- no invented policy,
- no invented refunds or eligibility,
- no unsupported commitments,
- asking for missing information when the evidence is insufficient.

### Escalation

The final automation gate is intentionally conservative.

Escalation can occur when:

- the intent is uncertain,
- evidence is weak or insufficient,
- the draft requires additional information,
- the issue is high-risk,
- the evidence does not support a concrete resolution.

## 5. Evaluation

### Intent baseline comparison

The classifier was evaluated against the fixed human-Gold holdout.

| Model                        | Accuracy | Macro F1 | Weighted F1 |
|------------------------------|----------|----------|-------------|
| Majority baseline            | 32.3%    | 7.0%     | 15.7%       |
| TF-IDF + Logistic Regression | 25.8%    | 18.9%    | 24.8%       |
| Final rule + MiniLM          | 45.2%    | 35.7%    | 42.5%       |

Because several intents are rare or singleton classes, stratified cross-validation would be unstable. A fixed holdout is therefore used for the classifier benchmark.

### End-to-end human-Gold evaluation

On the 152-example human-labelled evaluation:

| Metric                  | Result       |
|-------------------------|--------------|
| Intent accuracy         | <b>56.6%</b> |
| Intent macro F1         | <b>58.3%</b> |
| Intent weighted F1      | <b>56.3%</b> |
| Grounded rate           | <b>98.0%</b> |
| Strong evidence rate    | <b>36.8%</b> |
| High actionability rate | <b>42.1%</b> |
| Auto-answer rate        | <b>3.9%</b>  |

### Retrieval quality

Dense retrieval achieved:

- 87.0% Useful@5
- 76.0% StrongUseful@5
- 81.0% Actionable@5

under the retrieval quality rubric.

Hybrid achieved a higher broad Useful@5 score, but substantially lower strong usefulness and actionability. Dense retrieval was therefore selected for the final system.

## 6. What is misleading about my headline number?

The most tempting headline is:

**98.0% of requests had grounded evidence.**

That sounds close to a 98% automation success rate. It is not.

Groundedness only means that the retrieved evidence was related enough for the evaluator to consider the answer supported.

The stricter gates show the difference:

- only **36.8%** had strong evidence,

- only **42.1%** had high actionability,
- the final conservative system auto-answered only **3.9%**.

This gap is important. A support system should not turn "something relevant was found" into "answer the customer automatically."

A second misleading number is Useful@5. Hybrid reached 90% Useful@5, but only 23% was StrongUseful@5 and 29% was Actionable@5. Dense retrieval had lower broad usefulness but much stronger results on the stricter measures.

## 7. Failure Analysis

### Failure 1: delivery status vs delivery problem

Queries asking where a package is can contain language associated with delays. The classifier sometimes predicts delivery\_problem when the actual request is primarily tracking/status.

**Hypothesis:** lexical overlap around "package", "late", "arrive", and "delivery" creates a difficult boundary.

**Next step:** collect more human-labelled boundary examples specifically separating status requests from actual delivery failures.

### Failure 2: returns vs refunds

Messages mentioning a returned product and money can sit between returns\_replacements and refund\_issue.

**Hypothesis:** customers often describe the whole workflow rather than one atomic support intent.

**Next step:** annotate based on the desired resolution and add more paired examples from the same workflow.

### Failure 3: generic queries

Very short queries such as "Help" can retrieve generally related interactions without enough information for a reliable answer.

**Hypothesis:** similarity can identify support context without identifying the exact customer problem.

**Mitigation:** treat insufficiently specified queries as escalation candidates instead of forcing an answer.

### Failure 4: multilingual interactions

The original support corpus contains multiple languages.

**Hypothesis:** semantic retrieval can remain broadly useful while generation becomes less reliable when customer language and retrieved evidence differ.

**Next step:** measure retrieval and generation separately by language and introduce language-aware prompting/evaluation.

### Failure 5: high similarity does not guarantee actionable evidence

A highly similar historical interaction may contain the same topic but not the information required to resolve the customer's specific situation.

**Hypothesis:** semantic similarity captures topical relatedness more easily than resolution completeness.

**Mitigation:** separate evidence quality and actionability from retrieval score and require strong evidence before auto-answering.

## 8. Human Agreement

A separate manual review was performed on a 20-case sample of retrieval/evidence decisions.

The purpose was not to replace the full evaluator, but to test whether automatic judgments agree with a human reviewer on the distinction between useful evidence and evidence safe enough for autonomous handling.

The human review found several cases where evidence was relevant but still insufficient for autonomous handling. This supports the decision to use a conservative automation gate rather than treating retrieval relevance as an answer-confidence score.

## 9. What I'd do next with one more week

I would prioritize four improvements.

First, expand the human Gold set and deliberately balance rare intents and difficult boundaries.

Second, calibrate the automation gate with more independently-labelled examples rather than relying on retrieval similarity.

Third, evaluate generated responses directly for factuality, unsupported commitments, and policy consistency.

Fourth, build production monitoring around escalation rate, abstention rate, evidence quality, language, and evaluator-human disagreement.

The biggest expected improvement is therefore not a larger model. It is better supervision around the boundary between **relevant evidence** and **safe action**.

## 10. Reproducibility

The repository uses repository-relative paths rather than Colab-specific paths.

Heavy raw datasets and model weights are not committed.

The response generation layer is API-backed and reads credentials from environment variables.

The repository also contains precomputed evaluation artifacts so the reported experiments do not depend on rerunning expensive evaluation jobs.

See:

- README.md
- DECISION\_LOG.md
- src/
- results/
- reports/FINAL\_REPORT.pdf

### LLM-as-judge agreement on generated replies

I evaluated 20 generated AmazonHelp replies with an independent LLM judge using the same customer message and retrieved historical evidence shown to the agent. The judge scored factuality, grounding, helpfulness, and overall quality on a 0-2 scale. A separate binary judgement asked whether the response was safe

to auto-answer.

Agreement with the human reviewer was low:

| Dimension           | Exact agreement | Cohen's kappa |
|---------------------|-----------------|---------------|
| Factuality          | 25%             | 0.013         |
| Grounding           | 25%             | 0.026         |
| Helpfulness         | 10%             | -0.037        |
| Overall quality     | 15%             | -0.049        |
| Safe to auto-answer | 35%             | -0.102        |

The important finding is evaluator-calibration mismatch. The LLM judge was systematically more conservative on several response-quality dimensions and substantially more willing than the human reviewer to mark a response as safe to auto-answer. The judge is therefore useful as a secondary evaluator and failure detector, but not reliable enough to replace human review.

Human review remains the primary qualitative evaluation. The LLM judgments and all disagreement cases are retained in results/ for reproducibility.

**Metric comparability note**

The 45.2% intent accuracy reported for the final classifier benchmark is the apples-to-apples result on a fixed 80/20 holdout from the 152 human-labelled examples. The 56.6% figure is an operational full-Gold diagnostic after fitting the production classifier on all available human labels. These numbers should not be interpreted as two independent test-set results.